Introductory activity

Activity 1 · Turning and stretching a figure 40 min . We represent a point $M(x, y)$ by the column $X = \begin{pmatrix} x \\ y \end{pmatrix}$. For a matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, we define the product $AX = \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix}$: it is the column of a new point.

1. Let $R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Compute the images under R of the vertices of the triangle $O(0,0)$, $P(2,0)$, $Q(2,1)$. Draw both triangles. Which transformation do you recognise?
2. Let $E = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$. Same question.
3. Compute $R(EX)$ for any $X = \begin{pmatrix} x \\ y \end{pmatrix}$. Find a matrix C such that $R(EX) = CX$ for every X.
4. Compute likewise $E(RX)$ and the corresponding matrix. Are the two composite transformations the same? Check on the point Q.
5. Suggest a rule for computing C directly from R and E: it is the matrix product RE.

Observing is not proving: the properties conjectured here are established in the course.